

A Source-Weighting Mechanism to Enhance Evidence Quality for Fake News Detection

Hsin-Chun Tsai^{1*}, Jhing-Fa Wang², and Bo-Xuan Fang²

¹Department of Electrical Engineering Cheng Shiu University, Kaohsiung, Taiwan

²Department of Electrical Engineering, National Cheng-Kung University, Tainan, Taiwan

*Email: tsaihcmal@gmail.com

Abstract—The proliferation of online misinformation requires robust automated fact-checking systems. Many of these systems rely on evidence retrieved from search engines, making their accuracy highly dependent on the quality of the search results. This dependency creates a significant vulnerability, as low-quality, biased, or false information retrieved from the web can compromise the entire verification process. To address this “garbage in, garbage out” problem, we propose a Source-Weighting Mechanism designed to enhance the quality of evidence before it is used for final analysis. Our approach systematically triages and prioritizes information by implementing a multi-tiered, heuristic-based weighting system that assigns credibility scores to different source types. Highly credible sources, such as official fact-checking organizations and international news agencies, are assigned the highest weight. Reputable media outlets receive medium weight, while user-generated content from forums and unverified websites is given the lowest weight. By ranking and weighting evidence snippets according to source credibility, our mechanism mitigates the impact of informational bias and provides a more reliable foundation for downstream reasoning models. This work demonstrates a practical and effective method for improving the trustworthiness and accuracy of evidence-based fake news detection systems.

Index Terms—Fake News Detection, Evidence Retrieval, Source Credibility, Bias Mitigation, Fact-Checking, Information Trustworthiness, Evidence Quality

I. INTRODUCTION

The rapid proliferation of online social platforms has amplified the spread of multimodal fake news, posing a significant threat to societal trust and stability. In response, automated fact-checking systems have become crucial tools. Modern approaches increasingly leverage Large Language Models (LLMs), often through retrieval-augmented techniques where systems retrieve external evidence from web search engines to verify a claim’s veracity.

However, the efficacy of these retrieval-augmented systems is highly contingent on the quality of the evidence they retrieve. This dependency presents a critical vulnerability: if the search results are biased, contain misinformation, or are

from low-quality sources, the accuracy of the entire system can be severely compromised. Many existing systems treat all retrieved information sources with equal importance, failing to distinguish between credible journalistic reports and unverified user-generated content. This oversight can lead to the “garbage in, garbage out” problem, where even a sophisticated reasoning model produces unreliable verdicts because it is grounded in flawed evidence.

To address this fundamental challenge in evidence-based fact-checking, this paper introduces a Source-Weighting Mechanism. This mechanism functions as a critical pre-processing step designed to systematically evaluate and prioritize retrieved information based on the credibility of its source. Our approach implements a multi-tiered, heuristic-based weighting system that assigns different credibility scores to various source types, such as official fact-checking organizations, reputable news agencies, and online forums. By filtering and ranking evidence before the final analysis, our mechanism provides a more reliable foundation for downstream reasoning models.

II. RELATED WORK

A. Source Credibility and Evidence Quality Assessment

Evaluating information credibility is crucial in the digital age, with source credibility and content soundness emerging as key criteria [1]. These concepts can be articulated as authority/trustworthiness and plausibility/evidential support, respectively. The Internet’s accessibility has increased the importance of credibility assessment, as anyone can now publish content online [2]. To ensure transparency and reduce bias in systematic reviews, formal quality assessment tools are recommended over informal expert judgment [3]. These tools should demonstrate construct validity, facilitate inter-reviewer agreement, apply across study designs, and be user-friendly. When applying systematic review results to patient care, clinicians should evaluate the review’s credibility and the

quality of evidence [4]. This involves assessing the review’s methods, literature search, reproducibility, and presentation of results.

B. Stance Detection from Information Sources

Stance detection is a research area focused on determining an individual’s position towards a specific target from textual content, typically classifying stances as Favor, Against, or None [6]. This field has applications in social media analysis, trend forecasting, and election predictions [7]. Recent advancements include integrating contextual information through dual encoders with cross-attention, improving performance on both in-domain and out-of-domain targets [5]. Researchers are exploring various approaches, from historical methods to contemporary techniques, and developing datasets and tools to support stance detection research. The Stance Comp system exemplifies practical applications by aggregating stances from multiple sources, including social media and news outlets, to assist users in detecting potential rumors and verifying claims [8]. As stance detection continues to evolve, future research may expand to include image and video content analysis.

III. PROPOSED METHODS

To address the challenges of evidence quality and source credibility outlined in the previous section, we propose a systematic, two-stage mechanism. This mechanism is designed to operate after a complex claim has been broken down into a series of atomic, verifiable sub-questions. Its primary goal is to retrieve relevant evidence for each sub-question and, crucially, to weight this evidence based on the trustworthiness of its source before passing it to a downstream reasoning module.

A. Automated Evidence Retrieval

The first stage of our mechanism focuses on gathering a broad set of potential evidence. For each verifiable sub-question generated from the original claim, we transform it into a structured query. This query is then programmatically submitted to a web search engine via its API (e.g., Google Custom Search JSON API) to retrieve a list of relevant online sources. This process results in a collection of raw evidence snippets, each comprising a title, a brief description, and a source URL. At this stage, the evidence is unfiltered, reflecting the diverse and often unreliable nature of information available on the web.

B. The Heuristic-Based Source-Weighting System

The core innovation of our work lies in the second stage: a heuristic-based source-weighting system designed to systematically triage and prioritize the retrieved evidence. Challenging the implicit assumption that all web sources are equally reliable, we implement a multi-tiered weighting scheme that assigns a credibility score (w) to each piece of evidence based on its source domain. This system is structured as follows:

- **Highest Weight ($w = 1.0$):** This tier is reserved for sources with the highest standards of accuracy and impartiality. It includes established international news agencies (e.g., Reuters, Associated Press), dedicated fact-checking organizations (e.g., Snopes, PolitiFact, Taiwan Fact Check Center), and official academic or governmental domains (e.g., edu, gov). The rationale is that these sources employ rigorous editorial and verification processes.
- **Medium Weight ($w = 0.7$):** This tier includes reputable national and international media outlets (e.g., The New York Times, BBC News, Public Television Service in Taiwan). While these sources maintain professional journalistic standards, they may occasionally exhibit a specific editorial stance or report errors, thus warranting a slightly lower weight.
- **Lowest Weight ($w = 0.2$):** This tier is assigned to sources with low or unverified credibility. This primarily includes user-generated content (UGC) from online forums (e.g., PTT, Dcard) and anonymous or unverified personal websites. The information from these sources often represents personal opinion rather than fact and lacks professional oversight.
- **Default Weight ($w = 0.5$):** Any source that does not fall into the above categories is assigned a neutral default weight.

The output of this mechanism is a ranked and weighted list of evidence snippets for each sub-question. This curated set of information, which prioritizes high-quality sources, serves as a much more reliable foundation for any subsequent reasoning or verification model to make its final judgment.

IV. EXPERIMENTS

To validate the effectiveness and contribution of our proposed source-weighting mechanism, we conducted a series of experiments. The primary objective is to demonstrate that systematically evaluating and prioritizing evidence based on source credibility leads to a tangible improvement in fake news detection accuracy.

A. Experimental Setup

- **Dataset:** We conducted our primary evaluation on the **YTN-250 dataset**. This dataset consists of 250 news videos collected from various public YouTube channels, each manually verified and labeled as either “Factual” (125 videos) or “Erroneous” (125 videos). It was specifically created to test the system’s end-to-end performance on genuine video inputs in a real-world setting.
- **Evaluation Metrics:** We evaluated performance using three standard metrics: Precision, Recall, and F1-score. The F1-score, as the harmonic mean of precision and recall, is used as the primary metric for comparing model performance, especially in providing a balanced measure.

B. Ablation Study: Isolating the Mechanism’s Contribution

To specifically isolate the contribution of our evidence

quality enhancement process, we performed an ablation study on the YTN-250 dataset. We compared the performance of the full fact-checking model (which includes our source-weighting mechanism and subsequent verification) against a key variant:

- **w/o Similarity Verification:** In this variant, the structured, quantitative verification step is removed. Instead, the LLM is prompted to directly judge the consistency between the claim and the retrieved evidence, without the guidance of our mechanism’s credibility-weighted analysis. This allows us to measure the performance impact of our systematic evidence evaluation process.

C. Results and Discussion

The results of the ablation study are presented in Table I. The full model, which incorporates our source-weighting mechanism and the subsequent structured verification, achieves the highest F1-score of 84.29%.

In contrast, the “w/o Similarity Verification” variant, which relies on direct LLM judgment over the retrieved evidence, shows a notable performance drop, with its F1-score decreasing by 1.94% to 82.35%. This degradation highlights that simply retrieving a list of sources is insufficient. The structured process of evaluating source credibility and quantitatively measuring semantic consistency—core functions facilitated by our mechanism is less effective when removed.

These results validate that our source-weighting mechanism, as part of a systematic evidence evaluation pipeline, makes a significant and distinct contribution to the overall performance of the fake news detection system. It confirms the value of having a dedicated, fine-grained semantic check that operates on evidence curated for its quality and credibility.

TABLE I
ABLATION STUDY RESULTS ON THE YTN-250 DATASET

Experiment	P(%)	R(%)	F1(%)	$\Delta F1$
Full Model	88.00	80.88	84.29	-
w/o Similarity Verification	84.00	80.77	82.35	-1.94

V. CONCLUSION

This paper addressed the critical vulnerability of automated fact-checking systems, which are highly dependent on the quality of evidence retrieved from the web. To combat the “garbage in, garbage out” problem, we introduced a Source-Weighting Mechanism designed to systematically evaluate and prioritize retrieved information before the final analysis. Our approach implements a multi-tiered, heuristic-based system that assigns credibility scores to sources, giving the highest weight to official fact-checking organizations and international news agencies, medium weight to reputable media outlets, and the lowest weight to user-generated content. This mechanism ensures that downstream reasoning models are grounded in more reliable and trustworthy evidence.

The effectiveness of our proposed mechanism was validated through an ablation study on the YTN-250 dataset. The results

demonstrated that the full model, incorporating our source-weighting and verification process, achieved the highest F1-score of 84.29%. A variant of the model that did not use this systematic, credibility-weighted analysis saw its F1-score drop by 1.94% to 82.35%. This performance degradation confirms that our mechanism provides a significant and distinct contribution to the system’s overall accuracy. The findings validate that a dedicated process for curating evidence based on source credibility is a crucial step for improving the trustworthiness of evidence-based fake news detection systems. Future work could focus on expanding the granularity of source categories and exploring machine learning models to dynamically learn and assign source credibility scores.

REFERENCES

- [1] I. Vamanu and E. Zak, “Information source and content: articulating two key concepts for information evaluation,” *Inf. Learn. Sci.*, vol. 123, no. 1/2, pp. 65–79, 2022.
- [2] M. J. Metzger, “Making sense of credibility on the Web: Models for evaluating online information and recommendations for future research,” *J. Am. Soc. Inf. Sci. Technol.*, vol. 58, no. 13, pp. 2078–2091, 2007.
- [3] G. S. Bilotta, A. M. Milner, and I. L. Boyd, “Quality assessment tools for evidence from environmental science,” *Environ. Evid.*, vol. 3, no. 1, p. 14, 2014.
- [4] M. H. Murad, V. M. Montori, J. P. A. Ioannidis, R. Jaeschke, P. J. Devereaux, K. Prasad, et al., “How to read a systematic review and meta-analysis and apply the results to patient care: users’ guides to the medical literature,” *Jama*, vol. 312, no. 2, pp. 171–179, 2014.
- [5] T. Beck, A. Waldis, and I. Gurevych, “Contextual information integration for stance detection via cross-attention,” *arXiv preprint arXiv:2211.01874*, 2022.
- [6] D. Kućuk and F. Can, “A tutorial on stance detection,” in *Proc. 15th ACM Int. Conf. Web Search and Data Mining*, 2022, pp. 1626–1628.
- [7] D. Kućuk and F. Can, “Stance detection: Concepts, approaches, resources, and outstanding issues,” in *Proc. 44th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2021, pp. 2673–2676.
- [8] H. Xu and H. Fang, “StanceComp: Aggregating stances from multiple sources for rumor detection,” in *Asia Information Retrieval Symposium*, Springer, 2018, pp. 29–35.